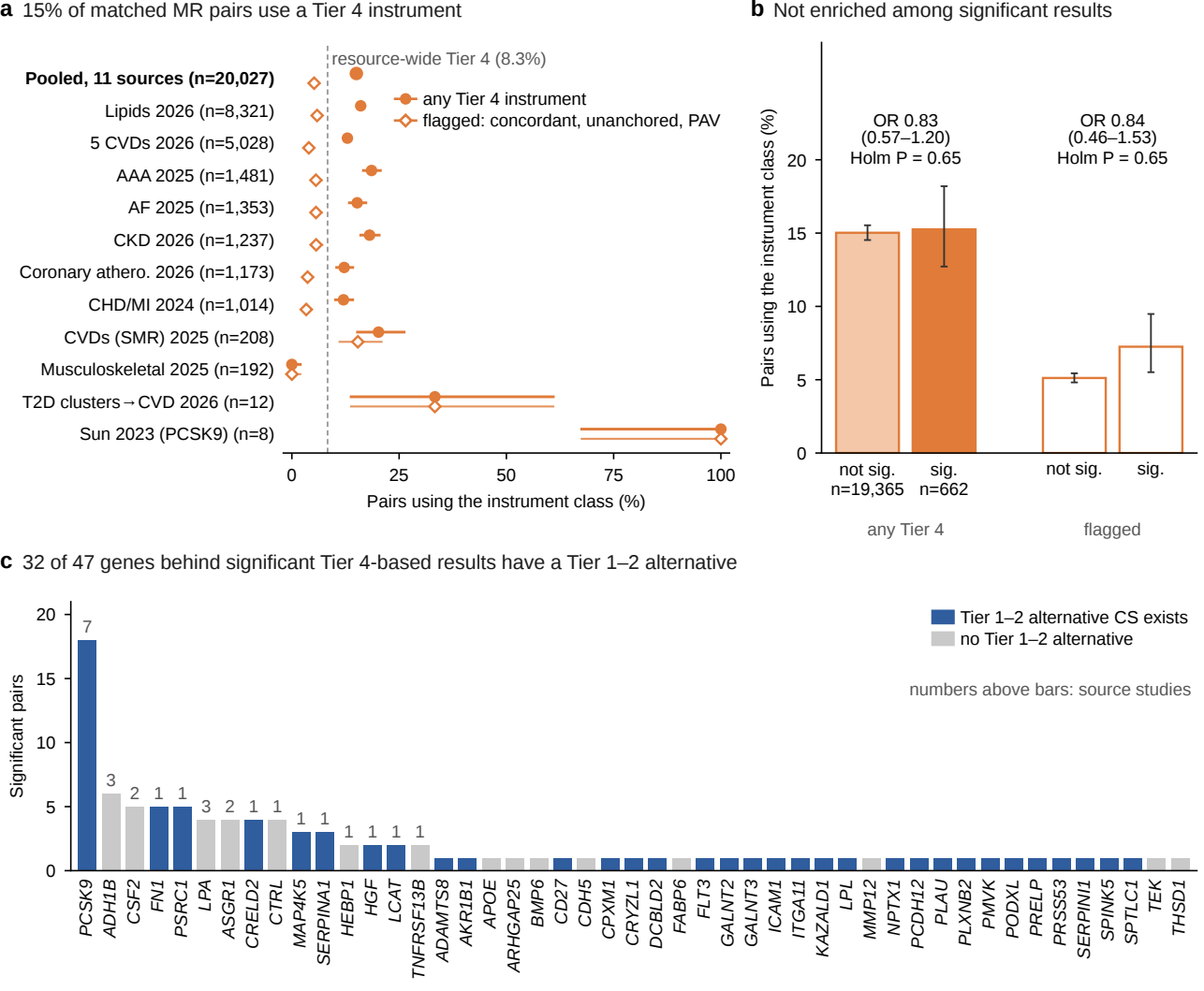
a 15% of matched MR pairs use a Tier 4 instrument
resource-wide Tier 4 (8.3%)
Pooled, 11 sources (n=20,027)
Lipids 2026 (n=8,321)
5 CVDs 2026 (n=5,028)
AAA 2025 (n=1,481)
AF 2025 (n=1,353)
CKD 2026 (n=1,237)
Coronary athero. 2026 (n=1,173)
CHD/MI 2024 (n=1,014)
CVDs (SMR) 2025 (n=208)
Musculoskeletal 2025 (n=192)
T2D clusters→CVD 2026 (n=12)
Sun 2023 (PCSK9) (n=8)
any Tier 4 instrument
flagged: concordant, unanchored, PAV
0
25
50
75
100
Pairs using the instrument class (%)
b Not enriched among significant results
Pairs using the instrument class (%)
0
5
10
15
20
OR 0.83
(0.57–1.20)
Holm P = 0.65
OR 0.84
(0.46–1.53)
Holm P = 0.65
not sig.
n=19,365
sig.
n=662
not sig.
sig.
any Tier 4
flagged
c 32 of 47 genes behind significant Tier 4-based results have a Tier 1–2 alternative
Significant pairs
0
5
10
15
20
Tier 1–2 alternative CS exists
no Tier 1–2 alternative
numbers above bars: source studies
7
3
2
1
1
3
2
1
1
1
1
1
1
1
1
PCSK9
ADH1B
CSF2
FN1
PSRC1
LPA
ASGR1
CRELD2
CTRL
MAP4K5
SERPINA1
HEBP1
HGF
LCAT
TNFRSF13B
ADAMTS8
AKR1B1
APOE
ARHGAP25
BMP6
CD27
CDH5
CPXM1
CRYZL1
DCBLD2
FABP6
FLT3
GALNT2
GALNT3
ICAM1
ITGA11
KAZALD1
LPL
MMP12
NPTX1
PCDH12
PLAU
PLXNB2
PMVK
PODXL
PRELP
PRSS53
SERPINI1
SPINK5
SPTLC1
TEK
THSD1